



CALCULUS BASED PHYSICS 2

SIR JOHN MICHAEL MENDOZA | MIDTERM | 1ST SEM

L1: WAVES AND SOUND

Nature of Waves

- ★ **Waves** - disturbance or oscillation that travels through matter or space accompanied by a transfer of energy
- ★ Two Basic Types of Waves:
 1. **Transverse Wave** - a type of wave in which the disturbance occurs perpendicular to the direction of travel of the wave (light)
 2. **Longitudinal Wave** - a type of wave in which the disturbance occurs parallel to the line of travel of the wave (sound)

PERIODIC WAVES

- A wave whose displacement has a periodic variation with time or distance, or both.
 - Transverse and longitudinal waves are classified as periodic waves.
- **Amplitude (A)** - maximum displacement moved by a point on a vibrating body or wave measured from its equilibrium position
 - Distance between crest and the undisturbed position; it is also the distance between a trough and the undisturbed position
 - **Crest** - highest point on the wave pattern
 - **Trough** - lowest point on the wave pattern
- **Wavelength (λ)** - horizontal length of one cycle of the wave (in meters)
- **Period (T)** - time required for one complete cycle (in seconds)
- **Frequency (f)** - number of cycles per second (in Hertz)

PERIODIC WAVE FORMULA

Frequency	$f = \frac{1}{T}$
Speed of any periodic wave	$v = \frac{\lambda}{T} = f\lambda$

WAVE ON A STRING

- The speed of a wave pulse traveling along a string or wire is determined by knowing its mass per unit length and its tension.

WAVE ON A STRING FORMULA

$v_{\text{wave of string}} = \sqrt{\frac{T}{m/L}} = \sqrt{\frac{TL}{m}}$	$T = \text{tension}$ $L = \text{length}$ $m = \text{mass}$
$f = \frac{v_{\text{wave of string}}}{\lambda} = \frac{v_{\text{wave of string}}}{2L} = \frac{1}{2L} \sqrt{\frac{TL}{m}}$	

NATURE OF SOUND

- **Sound** - a longitudinal wave created by a vibrating object; it can be created or transmitted only in a medium such as gas, liquid, or solid.
- **Pure Tone** - a single frequency sound
- **Infrasonic Waves** - waves with frequencies below 20Hz; too low for humans to hear
- **Sonic Waves** - waves with frequencies between 20Hz-20kHz; can be heard by the human ear
- **Ultrasonic Waves** - waves with frequencies above 20kHz; too high for humans to hear

CLASSIFICATION OF SOUND

- **Tones or Musical Sounds** - sound of one regular vibration or one being perceived by the ear to have musical sequence
- **Noise** - sound of irregular vibration or one with no definite range of frequency

CHARACTERISTICS OF SOUND

- **Loudness (amplitude)** - a physiological sensation that refers to the strength of the sensation as received by the ear; depends mainly on the sound amplitude
- **Intensity** - power carried by the wave through a unit area perpendicular to the direction of propagation
- **Pitch (frequency)** - attribute of auditory sensation in terms of which sound may be ordered on a scale primarily related to frequency
 - **Fundamental Frequency** - tone with lowest frequency
 - **Overtone** - tones other than the fundamental that produces the same note
 - **Harmonics** - overtones with frequencies that are integer multiple of the fundamental
- **Quality or Timbre (tone color)** - attribute of auditory sensation in terms of which a listener can judge that two sounds similarly presented and having the same loudness and different pitch

SOUND TERMINOLOGIES

- **Echo** - a reflected sound / repetition
- **Reverberation** - persistence of sound in space after the original sound is removed
- **Interference** - superimposition of one wave on another

SPEED OF SOUND FORMULA	
in air (T ≤ 20°C)	$V = 331.45 + 0.6T$ <i>V = velocity of sound in m/s</i> <i>T = temp of air in °C</i>
in air (T > 20°C)	$V = 331.45\sqrt{\frac{T_K}{273}}$ <i>V = velocity of sound in m/s</i> <i>T_K = temp of air in Kelvin</i>
in ideal gas	$V = \sqrt{\frac{\gamma RT}{MW}}$ <i>V = velocity of sound in m/s</i> <i>γ = ratio of specific heats; c_p/c_v = 1.4 for air</i>

	<i>R = gas constant, 8.314472 J/mol-K</i> <i>MW = molecular weight in kg/mol</i>
in other medium	$V = \sqrt{\frac{Y}{\rho}}$ <i>V = velocity of sound in m/s</i> <i>Y = modulus of elasticity</i> <i>ρ = density</i>

Note: The speed of sound is independent to pressure, frequency, and wavelength.

DOPPLER EFFECT

- The change in frequency or pitch of the sound detected by an observer because the sound source and the observer have different velocities with respect to the medium of sound propagation.

DOPPLER EFFECT FORMULA	
$f_o = f_s \frac{V \pm V_o}{V \mp V_s}$ <i>f_o = observed or apparent frequency in Hz</i> <i>f_s = frequency of the source in Hz</i> <i>v_o = observer's velocity in m/s</i> <i>v_s = source's velocity in m/s</i>	

L2: ELECTRIC FORCES AND ELECTRIC FIELD

Electric Charge

- ★ also known as charge, electrical charge, or electrostatic charge
- ★ characteristic of a unit of matter that expresses the extent to which it has more or fewer electrons than protons
 - **Proton** - positive charge
 - **Electron** - negative charge
 - **Neutron** - no net electric charge
- ★ the magnitude of the charge on the proton exactly equals the magnitude of the charge on the electron
- ★ most basic quantity in an electric circuit; electrical property of the atomic particles of which matter consists
- ★ SI unit: **Coulombs (C)**
 - charge *e* of an electron = $-1.602 \times 10^{-19}C$
 - charge of a proton = $1.602 \times 10^{-19}C$

Atom

- ★ consists of small relatively massive nucleus that contains particles called protons and neutrons
 - Proton Mass = $1.673 \times 10^{-27} kg$
 - Neutron Mass = $1.675 \times 10^{-27} kg$
- ★ surrounding the nucleus is a diffuse cloud of orbiting particles called electrons
 - Electron Mass = $9.11 \times 10^{-31} kg$

ELECTRIC CHARGE

- The coulomb is a large unit for charges.
 - $1\text{ C} = 1/(1.602 \times 10^{-19}) = 6.24 \times 10^{18}$ electrons
 - Thus, realistic or lab values of charges are on the order of: **pC**, **nC**, or **μC**
- The only charges that occur in nature are integral multiples of the electronic charge.
 - $e = -1.602 \times 10^{-19} C$
- The **law of conservation of charge** states that charge can neither be created nor destroyed, only transferred.
 - Thus, the algebraic sum of the electric charges in a system does not change
- Two electrically charged objects exert force on one another:
 - **Like charges** = repel each other
 - **Unlike charges** = attract each other
- Electric charge can move through an object.
 - **Electrical Conductors** - substances that readily conduct electric charge (ex: metals such as copper, aluminum, silver, gold – used in electrical wiring)
 - **Electrical Insulators** - materials that conduct electric charge poorly (ex: rubber, plastic, wood – coats electrical wiring)

COULOMB’S LAW

- The magnitude F of the electrostatic force exerted by one point charge q_1 on another point charge q_2 is directly proportional to the magnitudes q_1 and q_2 of the charges and inversely proportional to the square of the distance r between them:

COULOMB’S LAW FORMULA

$$F = k \frac{|q_1||q_2|}{r^2}$$

where k is a proportionality constant:
 $k = 8.99 \times 10^9 Nm^2/C^2$ in SI units

CENTRIFUGAL FORCE FORMULA

$$F_c = \frac{mv^2}{r}$$

m = mass
 v = velocity
 r = radius

ELECTRIC FIELD

- Electric Field ($\vec{E}$) that exists at a point is the electrostatic force $\vec{F}$ experienced by a small test charge q_0 placed at that point divided by the charge itself:

ELECTRIC FIELD FORMULA

$$\vec{E} = \frac{\vec{F}}{|q_0|}$$

- The electric field is a vector and its direction is the same as the direction of the force $\vec{F}$ on a positive test charge.
- SI Unit of Electric Field: **Newton per Coulomb (N/C)**